# TRANSITIVITY OF THE GALOIS GROUP ACTION: FOR A GALOIS CATEGORY (C, F) WITH G = AUT(F), THE ACTION OF G ON F(X) IS TRANSITIVE FOR EVERY CONNECTED X OF C.

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 747-755 of sources/stacks-pione/pinned-original-source.tex; proof span is lines 757-826, the complete author proof (68 body lines).

*Editorial note.* Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 13 of queue_pione.json (pb=3051, stmt 747-755, proof 757-826).

*Editorial note.* pione wave-1: fresh-adjudication accept (2026-10-14); label not covered by the 85 old-session S3-pi cards.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 00D3: sources/stacks-pione/pinned-original-source.tex, source lines 826–826. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 086J: sources/stacks-pione/pinned-original-source.tex, source lines 826–826. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0BMW: sources/stacks-pione/pinned-original-source.tex, source lines 565–575. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0BN1: sources/stacks-pione/pinned-original-source.tex, source lines 639–647. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0BN0: sources/stacks-pione/pinned-original-source.tex, source lines 624–647. Cited by the retained statement or proof but not

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part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0BN2: sources/stacks-pione/pinned-original-source.tex, source lines 705–708. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Compare with . Let $(\mathcal{C}, F)$ be a Galois category. Let $G = Aut(F)$ be as in Example 0BMW. For any connected $X$ in $\mathcal{C}$ the action of $G$ on $F(X)$ is transitive.*

*Proof.* We will use the results of Lemma 0BN0 without further mention. Let $I$ be the set of isomorphism classes of Galois objects in $\mathcal{C}$. For each $i \in I$ let $X_i$ be a representative of the isomorphism class. Choose $\gamma_i \in F(X_i)$ for each $i \in I$. We define a partial ordering on $I$ by setting $i \geq i'$ if and only if there is a morphism $f_{ii'} : X_i \to X_{i'}$. Given such a morphism we can post-compose by an automorphism $X_{i'} \to X_{i'}$ to assure that $F(f_{ii'})(\gamma_i) = \gamma_{i'}$. With this normalization the morphism $f_{ii'}$ is unique. Observe that $I$ is a directed partially ordered set: (Categories, Definition 00D3) if $i_1, i_2 \in I$ there exists a Galois object $Y$ and a morphism $Y \to X_{i_1} \times X_{i_2}$ by Lemma 0BN2 applied to a connected component of $X_{i_1} \times X_{i_2}$. Then $Y \cong X_i$ for some $i \in I$ and $i \geq i_1$, $i \geq I_2$.

We claim that the functor $F$ is isomorphic to the functor $F'$ which sends $X$ to

$$F'(X) = \operatorname{colim}_I \operatorname{Mor}_{\mathcal{C}}(X_i, X)$$

via the transformation of functors $t : F' \to F$ defined as follows: given $f : X_i \to X$ we set $t_X(f) = F(f)(\gamma_i)$. Using (0BN1) we find that $t_X$ is injective. To show surjectivity, let $\gamma \in F(X)$. Then we can immediately reduce to the case where $X$ is connected by the definition of a Galois category. Then we may assume $X$ is Galois by Lemma 0BN2. In this case $X$ is isomorphic to $X_i$ for some $i$ and we can choose the isomorphism $X_i \to X$ such that $\gamma_i$ maps to $\gamma$ (by definition of Galois objects). We conclude that $t$ is an isomorphism.

Set $A_i = \operatorname{Aut}(X_i)$. We claim that for $i \geq i'$ there is a canonical map $h_{ii'} : A_i \to A_{i'}$ such that for all $a \in A_i$ the diagram

$$\begin{array}{ccc} X_i & \xrightarrow[f_{ii'}]{} & X_{i'} \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle h_{ii'}(a)} \\ X_i & \xrightarrow{f_{ii'}} & X_{i'} \end{array}$$

commutes. Namely, just let $h_{ii'}(a) = a' : X_{i'} \to X_{i'}$ be the unique automorphism such that $F(a')(\gamma_{i'}) = F(f_{ii'} \circ a)(\gamma_i)$. As before this makes the diagram commute and moreover the choice is unique. It follows that $h_{i'i''} \circ h_{ii'} = h_{ii''}$ if $i \geq i' \geq i''$. Since $F(X_i) \to F(X_{i'})$ is surjective we see that $A_i \to A_{i'}$ is surjective. Taking the inverse limit we obtain a group

$$A = \lim_I A_i$$

This is a profinite group since the automorphism groups are finite. The map $A \to A_i$ is surjective for all $i$ by Categories, Lemma 086J.

Since elements of $A$ act on the inverse system $X_i$ we get an action of $A$ (on the right) on $F'$ by pre-composing. In other words, we get a homomorphism $A^{opp} \to G$. Since $A \to A_i$ is surjective we conclude that $G$ acts transitively on $F(X_i)$ for all $i$. Since every connected object is dominated by one of the $X_i$ we conclude the lemma is true. $\square$